



EDUNEXA

Design and Development of a Smart Platform for Skill-Based Online Tutoring

Capstone-I Project | Hybrid UG Programme in Computer Science, AI and Cybersecurity

Indian Institute of Technology Patna | 6 May 2026

Made with GAMMA



Project Overview

The Challenge

Traditional learning systems lack personalisation and real-time performance tracking, limiting effective learning outcomes. Rapid technological advancements have significantly increased demand for skill-based education.

Our Solution

EduNexa is a smart, skill-based online tutoring platform that connects students with qualified teachers through an efficient, structured digital interface. The platform bridges the gap between learners seeking skill development and educators possessing domain expertise.

Platform Features



Teacher Module

Educators register, showcase qualifications and expertise, create professional profiles, list courses or skills, and manage availability schedules.



Student Module

Learners register, access personalised dashboards, track academic progress, enrol in courses based on interests, and monitor skill development.



Smart Matching

Intelligent matching mechanism connects learners with skilled instructors, providing structured learning paths and real-time progress monitoring.

Technology Stack

1

Frontend Architecture

React.js with Next.js 14 App Router for dynamic, responsive user interface. Tailwind CSS for design system and Framer Motion for animations.

2

Backend Development

Node.js with Express framework for RESTful API development and server logic implementation.

3

Database System

Firebase Firestore for real-time data storage and management of user profiles and activities.

4

Authentication

JWT Tokens for secure session and access management across the platform.

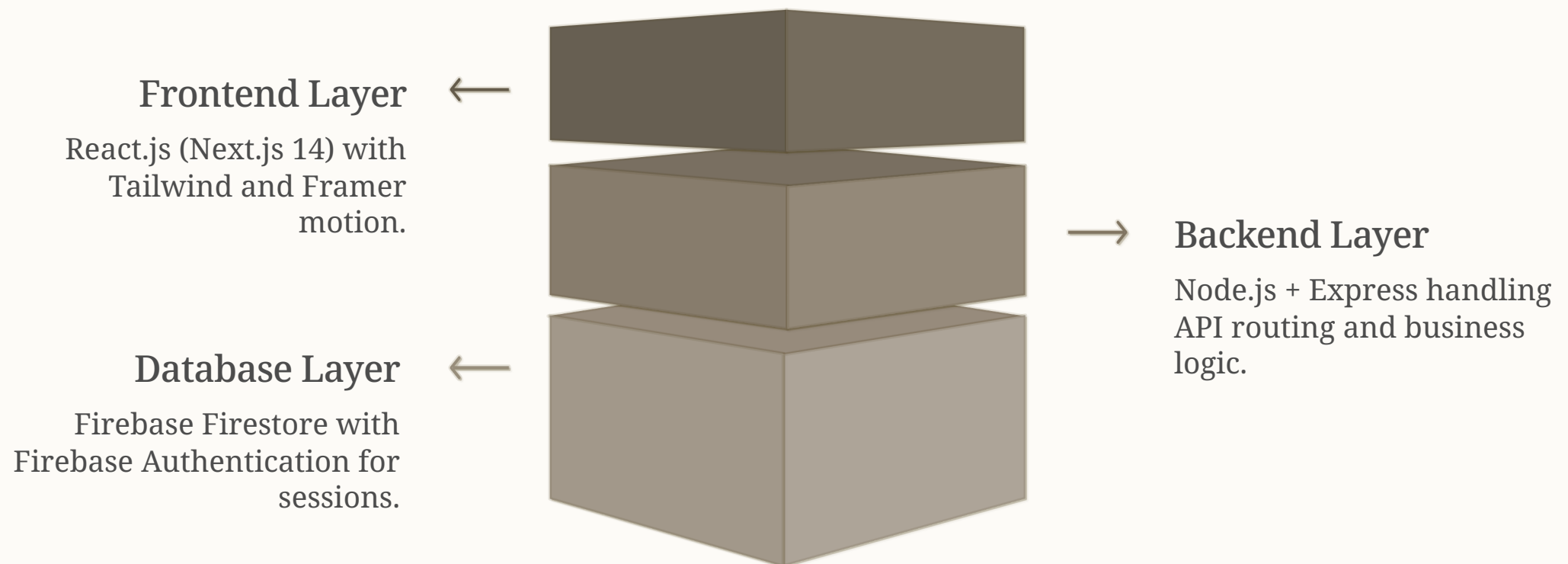


System Architecture

Client-Server Architecture

The platform follows a layered architecture with Next.js 14 App Router as the core framework, Firebase authentication for session management, and responsive UI components built with Tailwind and Framer Motion.

Data is stored in Firestore for real-time user profiles and activity tracking, whilst course catalog content is pre-loaded for quick access.



Project Structure



src/app/

Contains all page routes including authentication, student dashboard, and teacher portal



src/lib/

Firebase initialisation, authentication context, course data, and utility helpers



src/components/

Shared UI components such as Sidebar, DashboardLayout, and interactive elements



Configuration

Environment variables, Tailwind design system, and Next.js framework settings



Key Implementation Features



User Authentication

Secure JWT-based login and registration system with role-based access control



Personalised Dashboards

Dynamic interfaces tailored to student and teacher needs with real-time data



Progress Analytics

Interactive charts and performance tracking for continuous learning insights

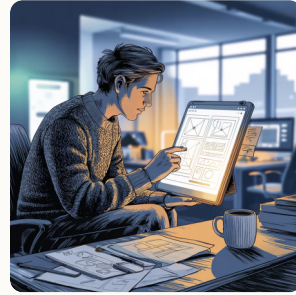
Team Contributions



Arun Kumar Giri

Project Lead & Full Stack Developer

Frontend, backend, database design, DevOps deployment, and integration



Ajeet Kumar Gautam

UI-UX Designer

s



Shailesh Kumar Giri

Quality Assurance



Amit Kumar Devanda

Documentation Lead



Aditya Kumar Dwivedi

Presenter

Future Enhancements



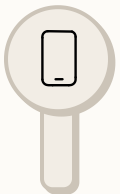
AI-Based Recommendations

Intelligent course suggestion system using machine learning algorithms to personalise learning paths



Live Video Integration

Real-time video classes with interactive whiteboard and screen sharing capabilities



Mobile Application

Cross-platform mobile apps for iOS and Android to enable learning on-the-go



Payment Gateway

Secure transaction system for premium courses and subscription-based services



Multi-Language Support

Regional language interfaces to make education accessible across diverse Indian demographics



Gamification

Achievement badges, leaderboards, and reward systems to boost student engagement

Conclusion

EduNexa demonstrates how modern technology can transform education by making it more accessible, personalised, and effective. The platform successfully integrates contemporary web technologies to deliver a smooth, responsive user experience whilst ensuring horizontal scalability for large user bases.

The system architecture supports seamless interaction between students and teachers, provides valuable insights into learning progress, and enables personalised education tailored to individual needs and goals.



Project Resources

Live Demo: <https://indian-institute-of-technology-caps-sigma.vercel.app/>

Repository: <https://github.com/ArPriCode/Indian-Institute-of-Technology-Capstone-Project-I.git>